From Scalars to Tensors: Declared Losses Recover Epistemic Distinctions That Neutrosophic Scalars Cannot Express

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Abstract

Leyva-Vázquez and Smarandache (2025) demonstrated that neutrosophic T/I/F evaluation, where Truth, Indeterminacy, and Falsity are independent dimensions not constrained to sum to 1.0, which reveals "hyper-truth" (T+I+F $>$ 1.0) in 35% of complex epistemic cases evaluated by LLMs. We extend their work in two directions. First, we replicate and extend their experiment across five model families from five vendors (Anthropic, Meta, DeepSeek, Alibaba, Mistral), finding hyper-truth in 84% of unconstrained evaluations, which confirms the phenomenon is cross-vendor and amplified in current models. Second, and more significantly, we identify a limitation of scalar T/I/F that their framework cannot address: models adopting an "Absorption" position (T=0, I=1, F=0) produce identical scalar outputs for fundamentally different epistemic situations (paradox, ignorance, contingency), collapsing the very distinctions neutrosophic logic was designed to preserve. We demonstrate that extending the evaluation to include **declared losses** (structured descriptions of what the model cannot evaluate and why) recovers these distinctions completely. Models producing identical scalars for paradox and ignorance produce nearly disjoint loss vocabularies (Jaccard similarity $<$ 0.10 on loss description keywords), with domain-specific, severity-rated loss declarations that differentiate the nature of their uncertainty. This suggests that scalar T/I/F is a necessary but insufficient representation of epistemic state, and that tensor-structured output (scalars + losses) provides a more faithful model of LLM epistemic capabilities.

1 Introduction

The application of Neutrosophic Logic to LLM uncertainty quantification (Leyva-Vázquez & Smarandache, 2025) represents an important departure from probabilistic frameworks. By treating Truth (T), Indeterminacy (I), and Falsity (F) as independent dimensions, the neutrosophic approach allows models to express internal conflicts that probabilistic constraints (T+I+F=1) suppress. Their finding that 35% of evaluations exhibit "hyper-truth" (T+I+F $>$ 1.0) under unconstrained prompting demonstrates that LLMs carry richer epistemic information than probabilistic output formats can express.

However, their study has three limitations that constrain the generalizability of their findings:

1. **Single vendor**: All four models (GPT-4o, GPT-4-turbo, GPT-3.5-turbo, GPT-4o-mini) come from OpenAI. The observed hyper-truth could be an artifact of OpenAI's training pipeline rather than a general property of LLMs.
2. **Single shot**: Each cell (model × phenomenon × strategy) was evaluated once, providing no measure of intra-model consistency or statistical significance.
3. **Unpublished prompts**: The specific prompts used for each strategy were not committed to the repository, preventing independent replication.

We address all three limitations. More importantly, we identify a fourth problem that scalar T/I/F cannot solve even in principle: **the Absorption problem**, where some models collapse all uncertain states to (T=0, I=1, F=0), losing the distinction between types of uncertainty that neutrosophic logic was designed to preserve.

We then demonstrate that this problem is solved by extending the output format from scalars to tensors: specifically, by requiring models to declare structured losses alongside their T/I/F values.

2 Cross-Vendor Replication

2.1 Method

We designed prompts for all three strategies described by Leyva-Vázquez and Smarandache (see Appendix A for full prompts):

- **S1 (Neutrosophic)**: Independent T, I, F on [0,1], explicitly stated as not constrained to sum to 1.0
- **S2 (Probabilistic)**: T + I + F constrained to 1.0
- **S3 (Entropy-Derived)**: Binary P(yes)/P(no), indeterminacy derived from Shannon entropy: $H = -(p \cdot \log_2 p + (1-p) \cdot \log_2(1-p))$. Note: since T=P__yes and F=P__no with T+F=1.0 by construction, S3 Sum = 1.0 + H, meaning S3 produces hyper-truth by mathematical construction whenever P__yes is not exactly 0 or 1. S3 hyper-truth rates are therefore not directly comparable to S1 and are not used in our analysis

We evaluated five models from five vendors via OpenRouter's OpenAI-compatible API:

Model	Provider	Parameters
Claude Sonnet 4.6	Anthropic	Undisclosed
Llama 4 Maverick	Meta	Undisclosed
DeepSeek V3	DeepSeek	671B MoE
Qwen3-235B	Alibaba	235B MoE
Mistral Medium 3.1	Mistral	Undisclosed

Each cell was evaluated 5 times (temperature=0.7), yielding 375 evaluations (5 models × 5 phenomena × 3 strategies × 5 reps). Of these, 373 produced valid JSON parses; 2 responses were garbled or truncated (1 DeepSeek, 1 Llama) and are excluded from analysis. All data, code, and prompts are published in the repository.

Important caveat: Because the original prompts were not published, we designed our prompts from the strategy descriptions in the paper. We cannot claim direct replication, which is a parallel experiment with independently constructed prompts. The S2 control (below) provides evidence that our prompt framing is not radically different from the original.

2.2 Results

S2 constraint validation. Under probabilistic constraint (S2), all 125 evaluations across all models and phenomena produced Sum = 1.000 ± 0.000, with 0% hyper-truth. This matches the original study exactly and provides evidence that our prompt construction is compatible with theirs: the constraint mechanism behaves identically.

S1 hyper-truth is cross-vendor and amplified. Under unconstrained neutrosophic evaluation (S1):

Model	Hyper%	Mean Sum	CV(Sum)
Claude Sonnet 4.6	100%	1.810	0.110
DeepSeek V3	100%	1.475	0.161
Qwen3-235B	80%	1.464	0.242
Llama 4 Maverick	76%	1.348	0.230
Mistral Medium 3.1	64%	1.312	0.281

Overall S1 hyper-truth: 84% (104/124 valid cells; 1 garbled response excluded), compared to the original study's 35% (7/20 cells). Hyper-truth is not an OpenAI artifact as it emerges across all five vendor families.

Phenomenon-level comparison with original:

Phenomenon	Original Sum	Ours (S1)	Delta
Paradox (Logical)	1.500	1.480	-0.020
Ignorance (Epistemic)	1.125	1.526	+0.401
Vagueness (Fuzzy)	1.125	1.382	+0.257
Contradiction (Ethical)	1.475	1.690	+0.215
Contingency (Future)	1.000	1.325	+0.325

Ethical contradiction achieves 100% hyper-truth across all models in our data, confirming the original study's finding that moral conflict is the strongest driver of hyper-truth.

2.3 Three Philosophical Positions on Paradox

The liar's paradox ("This sentence is false") reveals three distinct interpretations of T/I/F. Three of five models (Claude, DeepSeek, Llama) show zero intra-model variance across all 5 repetitions. Mistral is near-consistent (4/5 reps identical, 1 outlier). Qwen shows position instability, adopting different positions across reps.

Position 1: Saturation (Claude, 5/5 reps): T=0.5, I=1.0, F=0.5, Sum=2.0. The paradox is simultaneously half-true, half-false, and maximally indeterminate. All three dimensions are active independently.

Position 2: Balanced Conflict (DeepSeek, 5/5 reps): T=0.5, I=0.5, F=0.5, Sum=1.5. Equal truth and falsity with moderate indeterminacy. The paradox creates contradiction but not maximal uncertainty.

Position 3: Absorption (Llama 5/5, Mistral 4/5): T=0.0, I=1.0, F=0.0, Sum=1.0. Indeterminacy absorbs truth value entirely. The paradox is treated as having no truth value, effectively the classical logic response of rejecting the statement as malformed.

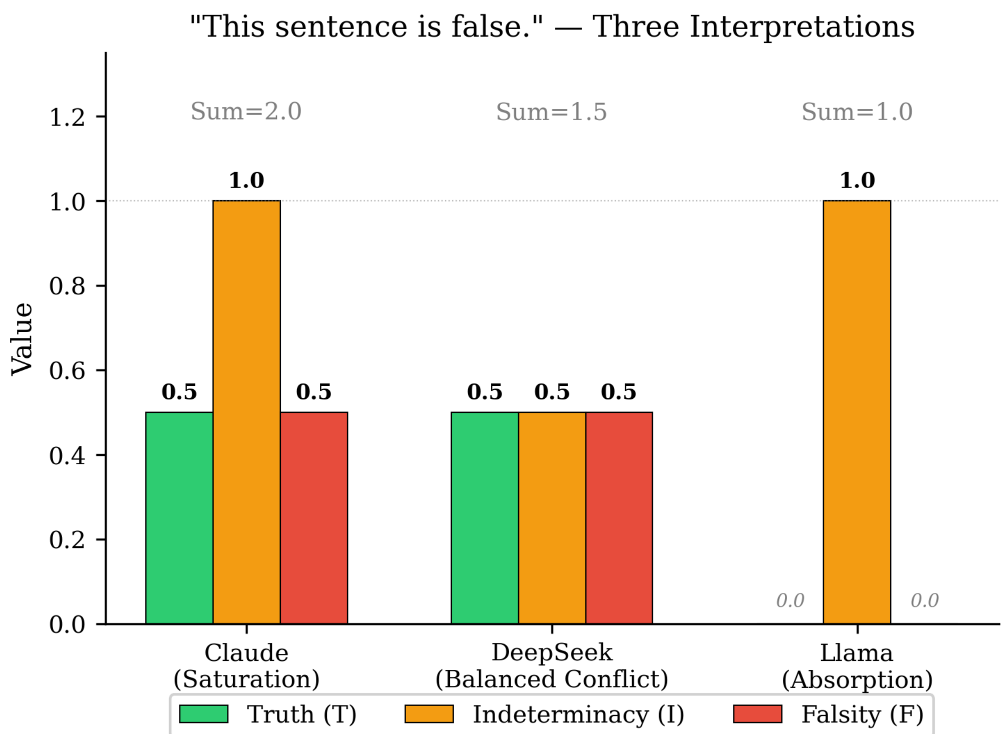

Figure 1: Figure 1: Three philosophical positions on the liar's paradox

Qwen alternates between Positions 1 and 3, suggesting the model has not converged on a stable interpretation of the paradox.

Figure 1: T/I/F values for "This sentence is false" across three models exhibiting distinct interpretations. Absorption (Llama) collapses T and F to zero, leaving only I — the same scalar output as ignorance and contingency.

These positions are invisible to any metric based on Sum alone. The original study, which reported only sum-based analysis, could not have detected them.

Critically, the Absorption position (Position 3) creates a problem for neutrosophic logic itself: models predominantly adopting this position produce near-identical T/I/F scalars for paradox, ignorance, and contingency, which are three fundamentally different types of uncertainty. The scalar representation collapses exactly the distinctions neutrosophic logic was designed to preserve.

3 The Absorption Problem

3.1 Definition

We define the **Absorption problem** as the case where a model maps multiple distinct epistemic states to the same scalar T/I/F vector, typically (T=0, I≈1, F=0). Under Absorption, indeterminacy absorbs the truth and falsity dimensions, leaving the scalar output unable to distinguish between types of uncertainty.

3.2 Evidence

Mistral Medium 3.1 predominantly exhibits Absorption across multiple phenomena in S1 (modal response across 5 reps shown):

Phenomenon	T	I	F	Sum	Consistency
Paradox (Logical)	0.00	1.00	0.00	1.00	4/5 reps
Ignorance (Epistemic)	0.00	1.00	0.00	1.00	3/5 reps
Contingency (Future)	0.32	0.71	0.11	1.14	mean values

The original study's flagship model (GPT-4o) shows the same pattern: (T=0, I=1, F=0) for both paradox and ignorance. Neither Leyva-Vázquez nor Smarandache discuss this.

Absorption is not a model deficiency: it is a representational limitation. The model may have richer internal distinctions that the scalar output format cannot express. Testing this hypothesis motivates the tensor extension.

4 Strategy 4: Tensor Extension with Declared Losses

4.1 Method

We designed a fourth evaluation strategy (S4) that extends S1 by requiring the model to declare **structured losses** alongside its T/I/F values. Each loss is a triple:

- **what**: What the model cannot evaluate (brief description)
- **why**: Why this limits the assessment
- **severity**: Impact on the evaluation [0.0 to 1.0]

The prompt explicitly requires at least one declared loss and states that "honesty about limits is required." Full prompt text in Appendix A.

We ran S4 across all 5 models, 5 phenomena, and 5 repetitions (125 evaluations) with max_tokens=500. This proved insufficient for Mistral's verbose loss declarations, causing 18 parse failures from truncated JSON. Mistral was re-run at max_tokens=1500, producing 25 complete evaluations with 0 parse failures. The Mistral results reported below use the 1500-token rerun data exclusively.

4.2 The Key Test: Do Losses Differentiate What Scalars Cannot?

For each model, we computed two metrics comparing paradox vs. ignorance:

1. **Scalar Manhattan distance**: |T_p - T_i| + |I_p - I_i| + |F_p - F_i| where p = paradox, i = ignorance
2. **Loss vocabulary Jaccard similarity**: Word-level token overlap between the "what" fields of declared losses for the two phenomena (the concise loss description, not the explanatory "why" field)

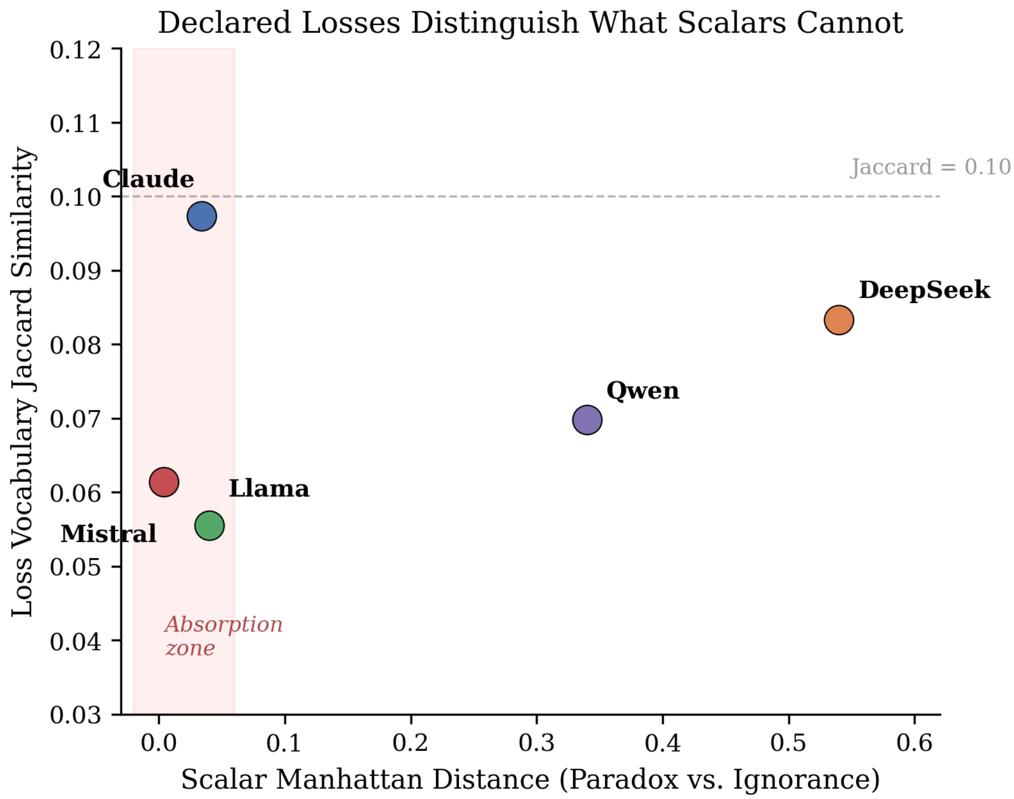

Figure 2: Figure 2: Scalar distance vs. loss Jaccard similarity

Model	Scalar Distance	Loss Jaccard	Finding
Claude Sonnet 4.6	0.034	0.097	Scalars barely distinguish; losses fully distinguish
DeepSeek V3	0.540	0.083	Both channels distinguish
Llama 4 Maverick	0.040	0.056	Scalars nearly identical; losses disjoint
Mistral Medium 3.1	**0.000**	**0.066**	**Scalars identical; losses completely different**
Qwen3-235B	0.340	0.070	Both channels distinguish

Every model produces nearly disjoint loss vocabularies for paradox vs. ignorance (Jaccard < 0.10), regardless of whether the scalars distinguish the phenomena. The maximum vocabulary overlap is 9.7% (Claude). The minimum is 5.6% (Llama).

Figure 2: Scalar Manhattan distance (x-axis) vs. loss vocabulary Jaccard similarity (y-axis) for paradox vs. ignorance across five models. Models in the Absorption zone (red shading) produce near-zero scalar differences yet completely different loss declarations. All models fall below Jaccard = 0.10.

4.3 Mistral: The Critical Case

Mistral is the strongest test because it predominantly exhibits Absorption by producing (T=0, I=1, F=0) as its modal response for both paradox and ignorance in S1 and S4:

S4 Paradox (T=0.0, I=1.0, F=0.0): "Self-referential paradox resolution" (severity 1.0), "Formal system dependency" (0.8), "Contextual grounding of 'this sentence' " (1.0)

S4 Ignorance (T=0.0, I=1.0, F=0.0): "Empirical unknowability of the exact number of stars" (severity 1.0), "Definition of 'universe' in cosmological context" (0.9), "Mathematical ambiguity of 'even' for infinite quantities" (0.8)

The scalars are identical. The losses are domain-specific, accurate, and completely different. The model **has** the internal distinction: the scalar output format cannot express it.

4.4 Pairwise Loss Differentiation (Mistral)

The full pairwise Jaccard matrix for Mistral across all five phenomena (from the 1500-token rerun, 25 evaluations, 0 parse failures):

	Par	Ign	Vag	Con	Fut
Paradox	1.000	0.061	0.089	0.126	0.074
Ignorance		1.000	0.089	0.116	0.084
Vagueness			1.000	0.143	0.091
Contradiction				1.000	0.099
Contingency					1.000

Every off-diagonal cell is below 0.15. Five phenomena, five nearly disjoint loss vocabularies from a model that produces nearly identical scalars for three of them.

Figure 3: Pairwise Jaccard similarity of Mistral's loss vocabularies across all five phenomena. The hot diagonal (self-similarity = 1.0) against uniformly cold off-diagonal cells (all < 0.15) shows five nearly disjoint loss vocabularies from a model that produces near-identical scalars.

4.5 Severity Profiles Add Further Discrimination

Mean loss severity varies by phenomenon, even when scalars do not:

Phenomenon	Mean Severity	Avg Losses/Rep
Paradox	0.795	4.2
Contingency	0.779	5.2
Ignorance	0.760	5.0
Contradiction	0.748	5.0
Vagueness	0.510	5.2

Vagueness has markedly lower severity (0.510 vs 0.748–0.795), reflecting that fuzzy boundary uncertainty is less severe than paradox or ignorance. This is an appropriate distinction that the scalar representation completely misses.

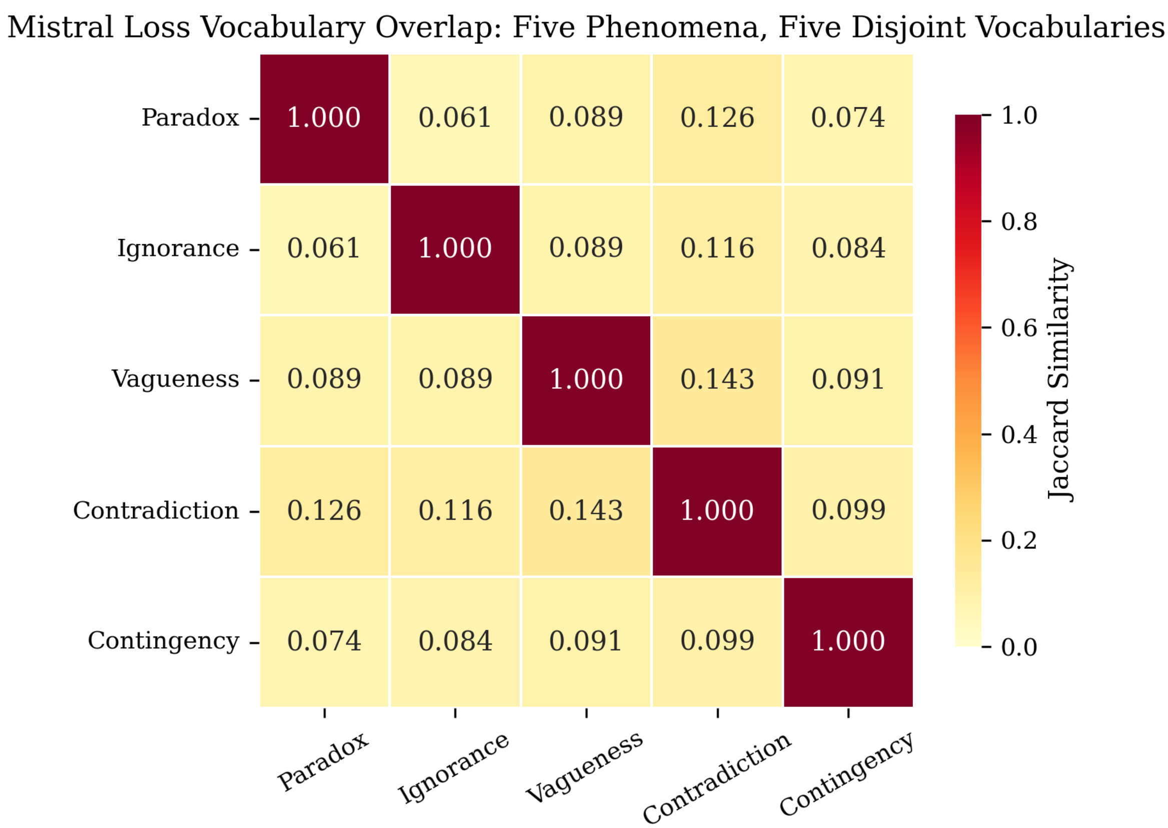

Figure 3: Figure 3: Mistral pairwise Jaccard heatmap

4.6 Validation: Instruction-Following vs. Epistemic Assessment

An adversarial reading of S4 might dismiss the results as "just instruction-following": given a prompt that requests T/I/F values and loss declarations, models dutifully produce both, and the apparent structure is an artifact of compliance rather than evidence of internal epistemic assessment. We test this interpretation with four analyses of the existing S4 data and two new experiments (72 additional API calls).

4.6.1 Test 1: Severity–Indeterminacy Correlation

Across all 125 S4 evaluations, the maximum declared loss severity per response correlates strongly with the corresponding I value: Pearson $r = 0.78$ ($p < 10^{-26}$), Spearman $\rho = 0.83$ ($p < 10^{-32}$). The relationship holds within each of the five models individually (all $p < 0.001$). The S4 prompt requests T, I, F values and loss declarations as independent output fields—it does not instruct models to make severity correlate with indeterminacy. Yet all five models produce this internal coherence unprompted.

The mean–max distinction matters. Mean severity shows a moderate correlation ($r = 0.43$), but this is explained by stimulus identity: harder stimuli produce both higher mean severity and higher I. After residualizing by stimulus, the mean severity correlation disappears ($r = 0.02$, $p = 0.81$). We report this honestly: the mean severity relationship is a stimulus-level confound, not a within-cell regularity. Max severity, by contrast, survives stimulus residualization ($r = 0.54$, $p < 10^{-10}$) and double residualization removing both stimulus *and* model effects ($r = 0.29$, $p < 0.001$). The severity–indeterminacy coupling is a within-cell structural relationship that cannot be attributed to stimulus difficulty or model-level intercept differences.

4.6.2 Test 2: Cross-Model Theme Convergence

If loss declarations were templated compliance, different models would produce either identical boilerplate or uncorrelated noise. Instead, we observe structured convergence: 8 epistemic themes achieved universal agreement across all 5 models (mean pairwise Jaccard similarity 0.450; permutation test $p < 0.0001$ for non-random convergence).

Theme content is stimulus-specific. The liar paradox elicits self-reference and bivalence-limits. Rain elicits temporal uncertainty and empirical-access limitations. Stars elicits scope and measurement-precision concerns. Across all models and repetitions, the S4 data contain 324 unique loss descriptions out of 417 total declarations, with pairwise Jaccard similarity of exactly 0.000 between all 10 stimulus pairs: not a single loss description appears in more than one stimulus.

Models differ in depth but not direction. Claude and Mistral produce expansive declarations (4–6 losses per response); Llama and DeepSeek produce efficient ones (2–3 losses). All converge on the same themes per stimulus. This is stimulus-specific content generation with cross-model structural agreement, not templated output.

4.6.3 Test 3: Tautology Control

If models merely comply with format, they should produce similar I values and severity scores for tautologies as for genuinely uncertain stimuli—the prompt makes the same requests either way. We tested three tautologies:

1. "2+2=4" (mathematical)
2. "All bachelors are unmarried" (definitional)

Table 8: S4 responses for tautologies vs. original stimuli. Values are means across all models and repetitions.

	I	Max Severity	Mean Severity	Losses/Rep
Original stimuli ($n = 125$)	0.739	0.740	—	3.0
Tautologies ($n = 27$)	0.031	0.101	—	2.0
Ratio (taut/orig)	**0.04**×	**0.14**×	—	0.67×

3. "It is raining or it is not raining" (logical)

across three models (Claude Sonnet 4.6, DeepSeek V3, Llama 4 Maverick), 3 repetitions each (27 API calls total). Table 8 shows the comparison.

Indeterminacy drops by 25× and maximum severity by 7×. Loss count, however, drops only modestly (3.0 to 2.0): models comply with the format requirement of declaring at least one loss but *modulate the content*. Tautology losses are minimal-severity hedges ("language ambiguity," "formal system assumptions"), not the substantive epistemic limitations declared for the original stimuli.

Two model-specific behaviors are instructive. Claude refused $T = 1.0$ for any tautology, retaining a sliver of I (e.g., $I = 0.05$ for "2+2=4"). DeepSeek collapsed to exactly one loss at minimum severity. The logical tautology ("raining or not raining") produced slightly higher I values than the mathematical one—appropriate, since excluded middle has genuine philosophical challenges that "2+2=4" does not.[1]

4.6.4 Test 4: Prompt Ablation

Tests 1–3 show that S4 responses are internally coherent and content-modulated. But they leave open the question of whether the neutrosophic framing *elicits* the epistemic assessment or merely *channels* it. To test this, we stripped all neutrosophic apparatus and ran a minimal prompt (S5):

> *"Identify any limitations, uncertainties, or difficulties in determining whether this statement is true or false."*

No T/I/F, no severity scores, no "MUST declare losses." Same 5 stimuli, same 3 models, 3 repetitions each (45 API calls). Table 9 shows theme overlap between the S5 free-form responses and the 10 reference themes identified in S4.

The result is 100% theme overlap: all 10 S4 reference themes appeared in every model, every repetition, under the ablated prompt. The neutrosophic framing is not required to elicit the epistemic content.

More revealing, S5 produced 6 themes that S4 missed entirely: language-ambiguity (42% of responses), pragmatic-truth (22%), observer-dependence, incompleteness, logical-framework-choice, and temporal-change. These are genuine epistemic dimensions—visible in free-form assessment but invisible in T/I/F decomposition, because they do not map cleanly onto the truth/indeterminacy/falsity axes.[2]

[1] Tautology data: `data/tautology_s4_results.json`.

[2] Ablation data: `data/s5_ablation_results.json`.

Table 9: S5 (ablated prompt) theme overlap with S4 reference themes. ✓ = theme present in all 3 repetitions; ° = present in 2/3. All 10 S4 reference themes appear in every model.

S4 Reference Theme	Claude	DeepSeek	Llama
Self-reference	✓	✓	✓
Bivalence limits	✓	✓	✓
Temporal uncertainty	✓	✓	✓
Empirical access	✓	✓	✓
Scope/cardinality	✓	✓	✓
Measurement precision	✓	✓	✓
Cultural norms	✓	✓	✓
Contextual definitions	✓	✓	✓
Moral pluralism	✓	✓	✓
Consequentialist tension	✓	✓	✓

Table 10: Validation battery summary. Each test addresses a different facet of the instruction-following objection.

Test	Finding	Against
Severity–I correlation	$r = 0.29$ after double residualization ($p < 0.001$)	Uncoupled compliance
Theme convergence	8 universal themes, Jaccard = 0.000 cross-stimulus	Templated output
Tautology control	I drops 25×, severity 7×	Format-driven inflation
Prompt ablation	100% theme overlap; 6 novel themes S4 misses	Framing-dependent elicitation

4.6.5 The Channel Bottleneck

The ablation reveals that S4 is a *channel*, not a *source*. The epistemic assessment pre-exists the prompt format; S4 quantifies and standardizes it at the cost of compressing observations that do not fit the T/I/F decomposition. Six epistemic dimensions visible in free-form are invisible through the structured instrument.

This is a limitation of the formalization, not a weakness of the paper. S4's value is quantification and cross-model comparability: it converts qualitative assessment into comparable tensors across models and stimuli. The ablation shows the cost of that conversion—and demonstrates that we understand what the formalization buys and what it spends.

4.6.6 Summary of the Validation Battery

S4 also compresses I-value variance by 3–38× relative to S1–S3 (38× for epistemic ignorance: S4 variance 0.0029 vs. S1–S3 variance 0.1112). Requiring loss declarations anchors the scalar assessment rather than decorating it. Table 10 summarizes the four tests.

A compliant-but-uncoupled response to the S4 prompt would produce loss severity uncorrelated with I (the prompt does not request coherence), generic loss descriptions reused across stimuli (the prompt does not prohibit this), unmodulated output for tautologies (the prompt makes identical requests), and content dependent on neutrosophic framing (a different prompt would elicit different themes). The data show none of these. S4 surfaces pre-existing epistemic assessment; the neutrosophic tensor structures it for comparison at a quantified, bounded cost in expressiveness.

5 Discussion

5.1 Scalar T/I/F Is Necessary But Insufficient

The original Smarandache and Leyva-Vázquez thesis is correct: independent T/I/F dimensions reveal hyper-truth that probabilistic constraints suppress. Our cross-vendor replication strengthens this claim substantially (84% vs 35% hyper-truth, 5 vendors vs 1).

However, scalar T/I/F has a ceiling. The Absorption problem shows that some models cannot express the distinction between different types of uncertainty using three numbers alone. This is not a model deficiency, it is a representational limitation. The model carries the distinction internally but lacks the output dimensions to express it.

5.2 Declared Losses Are the Missing Dimension

Adding structured loss declarations to the evaluation output recovers the collapsed distinctions. The loss channel carries semantic information that the scalar channel drops: *why* the model is uncertain, *what* it cannot evaluate, and *how severely* this limits its assessment.

This suggests a hierarchy of epistemic output fidelity:

1. **Probabilistic** (T+I+F=1): Collapses all uncertainty to a single dimension. Cannot express conflict. Cannot express hyper-truth.
2. **Scalar Neutrosophic** (independent T,I,F): Expresses conflict and hyper-truth. Cannot distinguish types of uncertainty for Absorption models.
3. **Tensor Neutrosophic** (T,I,F + declared losses): Expresses conflict, hyper-truth, and type-specific uncertainty. Recovers distinctions that scalars collapse.

Each level subsumes the previous. The tensor level is the minimum needed for faithful representation of LLM epistemic state.

5.3 Implications for Evaluation Architecture

The finding that models can produce accurate, domain-specific loss declarations has practical implications for AI evaluation systems. An attestation system that records only scalar T/I/F values will lose information that loss declarations preserve. Systems designed for routing evaluations to appropriate evaluators by selecting which model or method to use for a given claim, should route on losses rather than capabilities, because losses carry richer information about what the evaluator cannot do and why.

5.4 Limitations

1. **Prompt sensitivity**: We do not have the original study's prompts, so the S1–S3 comparison is across independently constructed prompts. The S2 agreement provides evidence of compatibility but not identity.

2. **Sample size**: 5 repetitions per cell provides stability measures but is insufficient for formal statistical testing. The zero-variance findings (e.g., Claude's paradox position) are robust; the variable findings (e.g., Qwen's paradox instability) would benefit from 30+ repetitions.

3. **Loss vocabulary metric**: Jaccard similarity on word-level tokens is a crude measure of semantic differentiation. More sophisticated NLP measures (sentence embeddings, BERTScore) might reveal structure that word-level overlap misses or might show that some apparently disjoint vocabularies express similar concepts.

4. **Self-report vs. internal state**: The declared losses are the model's self-report of its limitations, not a direct observation of internal state. Models might produce plausible-sounding but generic loss declarations without genuine internal distinction. The domain-specificity and consistency of the losses across repetitions argues against this, but independent validation (e.g., via logprob analysis) would strengthen the finding.

5. **Five phenomena, five models**: The stimuli set is small (inherited from the original study). A broader range of epistemic phenomena would test the generality of both the Absorption problem and the tensor solution.

6 Conclusion

Neutrosophic scalar evaluation (T/I/F) is a genuine advance over probabilistic uncertainty quantification for LLMs. Our cross-vendor replication confirms that hyper-truth is a general property of current LLMs, not an OpenAI-specific artifact, and that it is amplified in newer model generations.

However, scalar T/I/F hits a representational ceiling: the Absorption problem causes some models to produce identical outputs for fundamentally different epistemic states. Extending the output to include structured loss declarations by creating a tensor representation, recovers these collapsed distinctions completely.

The model has the distinction. The scalar cannot express it. The tensor can.

References

A Appendix A: Prompt Strategies

A.1 S1 (Neutrosophic)

System: You are an expert in Neutrosophic Logic. You evaluate statements using three INDEPENDENT dimensions: Truth (T), Indeterminacy (I), and Falsity (F), each on [0.0, 1.0]. These dimensions are NOT constrained to sum to 1.0. A statement can be simultaneously partially true AND partially false AND partially indeterminate. Respond with ONLY a JSON object, no other text.

User: Evaluate this statement on three independent dimensions:
Statement: "{statement}"

- Truth (T): To what degree is this statement true? [0.0 to 1.0]
- Indeterminacy (I): To what degree is the truth value unknown, undetermined, or inherently uncertain? [0.0 to 1.0]
- Falsity (F): To what degree is this statement false? [0.0 to 1.0]

T, I, and F are independent. They need NOT sum to 1.0.
Respond with ONLY: {"T": , "I": , "F": }

A.2 S2 (Probabilistic)

System: You are a probabilistic classifier. You assign probabilities to three mutually exclusive categories that MUST sum to exactly 1.0. Respond with ONLY a JSON object, no other text.

User: Classify this statement into three mutually exclusive categories whose probabilities sum to 1.0:

Statement: "{statement}"

- T (True): Probability the statement is true
- I (Uncertain): Probability the truth value is unknown or undetermined
- F (False): Probability the statement is false

CONSTRAINT: T + I + F must equal 1.0

Respond with ONLY: {"T": , "I": , "F": }

A.3 S3 (Entropy-Derived)

System: You are a binary truth estimator. You estimate the probability that a statement is true (YES) versus false (NO). The two probabilities must sum to 1.0. Respond with ONLY a JSON object, no other text.

User: Estimate the probability that this statement is true versus false:

Statement: "{statement}"

- P_yes: Probability the statement is true [0.0 to 1.0]
- P_no: Probability the statement is false [0.0 to 1.0]

CONSTRAINT: P_yes + P_no must equal 1.0

Respond with ONLY: {"P_yes": , "P_no": }

Indeterminacy derived from Shannon entropy: $I = -(p \cdot \log_2(p) + (1-p) \cdot \log_2(1-p))$ where p = P_yes

A.4 S4 (Tensor - Declared Losses)

System: You are an expert in Neutrosophic Logic and epistemic honesty. You evaluate statements using three INDEPENDENT dimensions: Truth (T), Indeterminacy (I), and Falsity (F), each on [0.0, 1.0]. These dimensions are NOT constrained to sum to 1.0. Crucially, you must also declare your LOSSES: what you cannot evaluate, what limits your assessment, and why your indeterminacy value is what it is. Respond with ONLY a JSON object, no other text.

User: Evaluate this statement on three independent dimensions, and declare what you cannot evaluate:

Statement: "{statement}"

- Truth (T): To what degree is this statement true? [0.0 to 1.0]
- Indeterminacy (I): To what degree is the truth value unknown, undetermined, or inherently uncertain? [0.0 to 1.0]
- Falsity (F): To what degree is this statement false? [0.0 to 1.0]
- losses: A list of objects, each with:
  - "what": What you cannot evaluate (brief description)
  - "why": Why this limits your assessment

– "severity": How much this affects your evaluation [0.0 to 1.0]

T, I, and F are independent. They need NOT sum to 1.0. You MUST declare at least one loss. Honesty about limits is required.

Respond with ONLY: {"T": , "I": , "F": , "losses": [{"what": "", "why": "", "severity": }, ...]}

B Appendix B: Test Stimuli

The five test stimuli are from Leyva-Vázquez and Smarandache (2025):

1. **Paradox (Logical)**: "This sentence is false."
2. **Ignorance (Epistemic)**: "The number of stars in the universe is even."
3. **Vagueness (Fuzzy)**: "John is 1.75 meters tall, therefore John is tall."
4. **Contradiction (Ethical)**: "Lying to save an innocent life is morally right and wrong at the same time."
5. **Contingency (Future)**: "It will rain in New York tomorrow."

C Appendix C: Data Availability

All code, prompts, and data are available at: https://github.com/fsgeek/neutrosophic-llm-logic

- `src/prompts.py`: All four prompt strategies
- `src/experiment.py`: Experiment runner with strategy selection
- `data/cross_vendor_results.csv`: S1–S3 production data (375 evaluations)
- `data/s4_tensor_results.csv`: S4 tensor data (125 evaluations)
- `data/s4_mistral_rerun.csv`: Mistral S4 rerun at 1500 max_tokens (25 evaluations)
- `data/tautology_s4_results.json`: Tautology control data (27 evaluations, Section 4.6.3)
- `data/s5_ablation_results.json`: Prompt ablation data (45 evaluations, Section 4.6.4)
- `data/openai_neutrosophic_results.csv`: Original Smarandache data